

Scheduling (reliable triggers)

Audience: operators / engineers

Chinese version: SCHEDULING.zh-TW.md (./SCHEDULING.zh-TW.md)

GitHub's native schedule cron is often delayed or skipped. This project uses:

1. Primary: external cron (cron-job.org) calling the GitHub API (workflow_dispatch)
2. Backup: sparse GitHub cron (Taiwan 10:30 / 15:30 / 18:30, weekdays)
3. Local (optional): Mac LaunchAgent (only when the Mac is on)
4. Manual: Actions UI or local scripts/trigger_update.sh

0. Current schedule overview (Taiwan time, Mon–Fri)

Layer	Source	Times	Notes
Primary	cron-job.org (https://console.cron-job.org/j)	09:10–16:10 hourly, 18:30	Jobs 8253638 (business hours), 8253639 (evening)
Backup	GitHub schedule	10:30, 15:30, 18:30	Fallback only; do not rely on it
Local	Mac LaunchAgent	same as primary	after ./scripts/install_launchd.sh;
Manual	local / Actions	anytime	Mac must be on ./scripts/trigger_update.sh or Run workflow

Monitor:

- Actions: <https://github.com/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions>
- cron-job.org: <https://console.cron-job.org/jobs>

0.1 What each run does

1. Download BOT CSV → convert to JSON
2. Write data/bot-xrt-latest.*; archives under data/history/ (90-day retention)
3. Rates changed → commit / push and purge jsDelivr CDN
4. Rates unchanged → workflow still succeeds but no new commit (expected)

Check changed=true/false in logs, or whether fetchedAtUtc refreshed in JSON.

0.2 Schedule not running? (step-by-step)

Step 1: GitHub Actions

Open Actions (<https://github.com/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions>).

Observation	Meaning
workflow_dispatch + success	Trigger works

No new runs	continue to Step 2
Runs but failure	inspect download/commit logs

Step 2: cron-job.org

Open Jobs (<https://console.cron-job.org/jobs>) → BOT FX Rates → History.

HTTP	Meaning
204	dispatch OK
401 / 403	GitHub token expired or missing Actions: write → rerun setup_cron_job.py
404	wrong workflow URL
No history	job disabled or not yet due

Recreate jobs:

```
export CRONJOB_API_KEY="your_api_key"
python3 scripts/setup_cron_job.py
```

Step 3: Manual trigger (fastest test)

```
./scripts/trigger_update.sh
```

Or Actions → Update BOT FX Rates → Run workflow.

Manual OK + auto fails → cron-job.org or token issue, not the download script.

Step 4: Local Mac fallback (optional)

```
./scripts/install_launchd.sh
```

Logs: /tmp/bot-fx-trigger.log, /tmp/bot-fx-trigger.stderr.log

0.3 Common misconceptions

Symptom	Reality
Actions ran, no new commit	rates unchanged (changed=false) — normal
CDN JSON looks the same	same reason; or jsDelivr cache — purge if needed
Nothing on weekends	by design (weekdays only)
GitHub schedule skipped	known issue; rely on cron-job.org
Public holidays still run	cron excludes weekends only, not TW holidays

0.4 Recommended weekly checks

1. cron-job.org History shows recent 204 responses
2. GitHub Actions has successful weekday workflow_dispatch runs

3. CDN JSON fetchedAtUtc is recent:

<https://cdn.jsdelivr.net/gh/ChesleyLo/DownloadBankOfTaiwanExchangeRate@main/data/bot-xrt-latest.json>

0.5 How to confirm the schedule ran (day-to-day)

Check in this order, fastest first:

① GitHub Actions (~30 seconds, start here)

Open: <https://github.com/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions>

Find Update BOT FX Rates:

What you see	Meaning
New workflow_dispatch with green ✓	Schedule triggered; run succeeded
No new runs	Schedule may not have fired → go to ②
Run exists but red ✗	Triggered but failed → open logs

② cron-job.org (confirm external cron hit GitHub)

Open: <https://console.cron-job.org/jobs>

Open BOT FX Rates - Business hours or Evening → History:

HTTP	Meaning
204	Successfully dispatched GitHub workflow
401 / 403	GitHub token expired → rerun setup_cron_job.py
No entries	Job disabled, or not yet due

③ CDN JSON (check data timestamp)

Open:

<https://cdn.jsdelivr.net/gh/ChesleyLo/DownloadBankOfTaiwanExchangeRate@main/data/bot-xrt-latest.json>

Check fetchedAtUtc:

- Recent → data was fetched
- Stale → may not have run, or rates unchanged with no commit (confirm with ①)

fetchedAtUtc updates only when a commit is pushed. If Actions ran with changed=false, CDN time may stay old — that is normal.

④ Manual test (when in doubt)

```
./scripts/trigger_update.sh
```

Or Actions → Run workflow

Outcome	Meaning
Manual OK, auto missing	cron-job.org issue, not the download script
Manual also fails	inspect Actions logs

⑤ Weekly checklist

- [] Weekday workflow_dispatch successes in Actions
- [] cron-job.org History shows 204
- [] (optional) CDN fetchedAtUtc looks reasonable

Full triage when nothing runs: §0.2.

1. Recommended external cron (cron-job.org, free)

1.1 Create a GitHub token

1. GitHub → Settings → Developer settings → Personal access tokens
2. Prefer a fine-grained token:
 - Repository access: only DownloadBankOfTaiwanExchangeRate
 - Permissions: Actions: Read and write
1. Copy the token (shown once)

Never commit the token. Store it only in the external cron service's secret fields.

1.2 One-command setup (when you have an API key)

1. Sign up at cron-job.org (<https://cron-job.org>) → copy API Key from Settings
2. Run locally (GitHub token uses gh auth token automatically):

```
export CRONJOB_API_KEY="your_cron_job_api_key"
python3 scripts/setup_cron_job.py
```

Creates two weekday jobs:

- 09:10–16:10 hourly
- 18:30 evening catch-up

Preview without creating:

```
python3 scripts/setup_cron_job.py --dry-run
```

1.3 Manual setup on cron-job.org (optional)

If you prefer the web UI:

Field	Value
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Title	BOT FX Rates - Update
URL	https://api.github.com/repos/ChesleyLo/DownloadBankOfTaiwanExchangeRate/actions/workflows/update-rates.yml/dispatch
Schedule	See below
Request method	POST
Timezone	Asia/Taipei

Headers:

```
Accept: application/vnd.github+json
Authorization: Bearer <YOUR_GITHUB_TOKEN>
X-GitHub-API-Version: 2022-11-28
Content-Type: application/json
```

Body:

```
{"ref":"main"}
```

1.4 Suggested times (Taiwan)

09:10, 10:10, 11:10, 12:10, 13:10, 14:10, 15:10, 16:10, 18:30

Custom cron (if supported), timezone Asia/Taipei:

```
10 9-16 * * 1-5
30 18 * * 1-5
```

1.5 Success signals

- cron-job.org shows HTTP 204
- GitHub Actions shows a workflow_dispatch run for Update BOT FX Rates

2. Local Mac fallback (installable)

If your Mac is usually on, install a LaunchAgent (uses gh auth, no extra token):

```
./scripts/install_launchd.sh
```

Runs on weekdays at 09:10–16:10 and 18:30. Log: /tmp/bot-fx-trigger.log

3. Local / server crontab

```
# A) gh already logged in
./scripts/trigger_update.sh

# B) PAT
export GITHUB_TOKEN=ghp_xxx
./scripts/trigger_update.sh

# C) repository_dispatch
```

```
./scripts/trigger_update.sh --mode repository_dispatch
```

Example crontab:

```
10 9-16 * * 1-5 GITHUB_TOKEN=ghp_xxx /path/to/repo/scripts/trigger_update.sh >>/tmp/bot-fx.log 2>&1
30 18 * * 1-5 GITHUB_TOKEN=ghp_xxx /path/to/repo/scripts/trigger_update.sh >>/tmp/bot-fx.log 2>&1
```

4. GitHub backup schedule

UTC	Taiwan	Role
30 2 * * 1-5	10:30	backup
30 7 * * 1-5	15:30	backup
30 10 * * 1-5	18:30	backup

Treat the external cron as primary.

concurrency.group: update-bot-rates prevents overlapping runs from fighting each other.

5. Troubleshooting (quick reference)

Symptom	Action
401/403 from cron	GitHub token expired or missing Actions: write; rerun setup_cron_job.py
404	wrong workflow file / repo path
204 but no Actions run	workflow disabled, or wrong ref in body
Run OK but no commit	rates unchanged (changed=false) — normal
No runs at all	follow §0.2 step-by-step
Stale CDN data	check recent commits; purge jsDelivr if needed

Full triage flow: §0.2 Schedule not running?

6. Related files

- .github/workflows/update-rates.yml
- scripts/trigger_update.sh
- scripts/setup_cron_job.py (one-click cron-job.org setup)
- scripts/install_launchd.sh (Mac local fallback)
- TECHNICAL.en.md (./TECHNICAL.en.md)